The Verification Gap: Checking the Reasoning in AI-Assisted Academic Writing

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paper1-draft-1-40-g9b6e252-dirty

Abstract

Generative AI tools now help write a growing share of academic papers. Checking whether their citations exist has received attention, but a paper whose contribution is an argument can cite only real sources and still reason badly. Reporting guidelines appear to offer little help here. The EQUATOR (Enhancing the QUAlity and Transparency Of health Research) Network lists about 700 guidelines for empirical health research, and I found no consensus checklist for verifying the reasoning of non-empirical papers. This commentary argues that the vocabulary for such a checklist already exists. Toulmin's layout of arguments treats every assertion as a claim that rests on grounds through a warrant and carries a qualifier; Whetten's criteria judge a theoretical contribution as a whole. Combined, they allow a procedure that I propose: register each claim with what it rests on, check each item by its kind, and check the contribution once. The paper applies this procedure to itself and publishes the record. It reports no evaluation; whether the procedure helps authors and reviewers is an open question.

Keywords: academic writing, generative AI, verification, argumentation, reporting guidelines, non-empirical research

Key Points:

- Generative AI tools can produce fluent papers whose citations are real and whose reasoning is still weak.
- Reporting guidelines cover empirical research; I found no consensus checklist for verifying the reasoning of non-empirical papers.
- Toulmin and Whetten supply a vocabulary for such a checklist: each claim is checked by what it rests on, and the contribution is checked once.
- The paper publishes its own claim registry as a worked example; it does not show that the procedure works.

1 Introduction

Writing assistants based on large language models (LLMs) have spread quickly: up to 22% of computer science papers now show signs of LLM editing, and such editing can be detected in every major discipline (Liang et al., 2025). In this paper "generative AI tools", or "AI" for short, means these assistants. They write fluently, and they can synthesise literature, structure an argument, draft prose and format references. When a human author falls short, the text tends to show it, in awkward phrasing, gaps in coverage or inconsistent formatting. An AI-assisted draft can read as polished and complete while it contains errors that only checking would reveal, such as a citation to a paper that does not exist.

Citation hallucination is the most discussed of these errors. Mugaanyi et al. (2024) found that 62–89% of the digital object identifiers (DOIs) that LLMs generated for natural-science and humanities topics were fabricated, and Walters and Wilder (2023) found fabricated citations in 55% of references generated by OpenAI's GPT-3.5 model and 18% of those by GPT-4 in generated literature reviews. A fabricated citation cannot be caught by knowing the field, because the work it cites does not exist; it can be caught by looking the work up.

A second error is harder to see. I call it *confidence inflation*: wording that expresses more certainty than the evidence supports. The evidence indicates that LLMs verbalise high confidence even on questions they answer wrongly (Xiong et al., 2024; Zhou et al., 2024) and that their summaries of scientific work generalise findings beyond what the original studies claim (Peters and Chin-Yee, 2025). Whether this carries over into academic prose is unsettled: in one corpus of applied-linguistics abstracts, ChatGPT used boosters such as "shows" at twice the human rate, but also more hedges (El-Dakhs et al., 2025). The stakes are epistemic, because words such as "demonstrates" and "suggests" are meant to tell a reader how much weight a claim can bear.

In technical papers a third error warrants investigation: derived numbers that look plausible and are wrong. In informal trials on my own projects, prompting a model to recompute every value caught arithmetic errors that prompting it to judge soundness

missed; this is an observation, not a measured result.

Recent systems go beyond checking that references exist. A linter checks a manuscript's references, internal consistency and causal language against the study design (Samsonau, 2026); a workflow pairs each drafted claim with an evidence packet and returns unsupported claims to the author (Chen et al., 2026); a provenance discipline binds reported quantities to an audit trail (Zhou and Yu, 2026); and a checklist for verifying cited assertions in narrative reviews is in development (VANRA Development Group, 2026). Each asks whether evidence exists, supports a claim or suffices, and the linter also checks causal wording against the study design. None that I found asks whether a warrant holds for its readers, or chooses the check by what a claim rests on, which is where a paper whose contribution is an argument can fail.

Detecting AI-written text does not fill that gap. Detection asks who wrote a text; verification asks whether it holds. Detectors have also proved fragile in tests: their accuracy varies with generation settings (Dugan et al., 2024), and paraphrase can collapse it (Krishna et al., 2023).

2 Reporting Guidelines and Non-Empirical Papers

A well-developed example of verification infrastructure for academic writing is the EQUATOR Network, which lists about 700 reporting guidelines for health research (EQUATOR Network, n.d.). Guidelines such as CONSORT for randomised trials and PRISMA for systematic reviews (Page et al., 2021) give authors a checklist to complete while they prepare a manuscript, and the GoodReports tool picks the checklist that fits a study type (Struthers et al., 2021). For one guideline the effect has been measured: journals endorsing CONSORT reported more completely on 25 of 27 checklist items, significantly so for five (Turner et al., 2012), and incomplete reporting is a recognised source of research waste (Glasziou et al., 2014).

Guidance for papers without new data exists, but it is scattered and seems not to reach the reasoning. Conceptual papers lack "commonly accepted templates" (Jaakkola, 2020, p. 18), although Jaakkola proposes four. Design-science researchers have reported

no agreed guidance on presenting their work (Gregor and Hevner, 2013; Dinter and Krawatzeck, 2015), and validity in that field "is neither well-understood nor consistently established" (Larsen et al., 2025). Software engineering's community standards ask methodological papers for "clear, valid arguments" but give no procedure for checking them (Ralph et al., 2020). A quality scale for narrative reviews exists, though its authors call it "not a reporting guideline" (Baethge et al., 2019), and reporting items for concept analysis have only recently been collected (Nashwan et al., 2026). The nearest neighbour is JBI's consensus-based checklist for appraising expert opinion, which asks "Does the opinion demonstrate a logically defended argument to support the conclusions drawn?" and refers reviewers to Toulmin's model (McArthur et al., 2025; JBI, 2026). It is a tool for reviewers who decide whether to include an opinion in a systematic review, and it asks one question about the argument as a whole. A documented search of the EQUATOR library and four bibliographic databases in October 2026 found no consensus checklist for verifying the reasoning of non-empirical papers step by step: whether each step from evidence to claim holds.[1]

Gregor (2006) distinguishes five types of theory in information systems research. Two of them, analytic theories (Type I) and theories for design and action (Type V), contribute conceptual structures more than findings, and such contributions may not be checkable against an earlier source, because they are the new contribution. Theoretical, conceptual, design-science and perspective papers are examples of non-empirical papers in this sense; the list is not a classification.

Building on Toulmin (2003) and Whetten (1989), I suggest that for these papers the main question may be less "do the data support the claim?" and more "does the reasoning hold?". That would be a different task from the one reporting guidelines support.

[1]Protocol, queries and screened candidates are in the repository (see Code and Data Availability), so the search can be repeated.

3 Toulmin and Whetten as a Basis for Verification

Two well-known frameworks may offer much of what a checklist for reasoning needs. Box 1 gives the terms; the framework's glossary defines them from their sources.[2]

Box 1. Terms used in this paper.

Term	Meaning
Claim	Any assertion put forward for acceptance.
Grounds	What a claim rests on (Toulmin's *data*): a source, other claims, a declared starting point, or the author's own observation.
Warrant	The step from grounds to claim.
Qualifier	The hedge that states how strong the claim is.
Confidence tier	The rating (Established, Supported, Emerging, Speculative) that decides which qualifiers a claim may carry.
Priority	How much the paper depends on a claim (P0 to P2).
Rebuttal	Circumstances in which the step from grounds to claim no longer holds.
Declared starting point	An assumption, definition or value the paper adopts without arguing for it.
Contribution	What the paper adds; checked once, as a whole.
Verification	Checking each claim against its grounds and warrant.
Validation	Checking that the contribution serves its readers and purpose.

Toulmin modelled his layout of arguments on jurisprudence. He called logic "generalised jurisprudence" and likened the difference between data and warrant to "the distinction drawn in the law-courts between questions of fact and questions of law" (Toulmin, 2003, pp. 7, 92). In his account any assertion is a claim; its *data* are "the facts we appeal to as a foundation for the claim"; its warrant is one of the "general, hypothetical statements, which can act as bridges" from data to claim; a *qualifier* indicates "the strength conferred by the warrant on this step"; and a *rebuttal* names "circumstances in which the general authority of the warrant would have to be set aside" (Toulmin, 2003, pp. 90–94). Toulmin later wrote that he had not had in mind an analytical model like the one that came to be called "the Toulmin model"; his concern "was with twentieth-century epistemology, not informal logic" (Toulmin, 2003, p. vii). Using his layout as a checklist is my adaptation, and it has to be judged on its own merits.

That adaptation looks workable. Two studies of university students' essays scored argument quality element by element on Toulmin's layout, with two raters each (Yang, 2022; Divya and Alamelu, 2026), which indicates that the elements can serve as assessment

[2]The glossary, the claim registry and the checking tools are public; see Code and Data Availability.

criteria; the second study adapted the first one's rubric, and both concern student essays, so applying the elements to verify research papers is a further step. LLMs can also extract Toulmin's elements from text when prompted with them (Gupta et al., 2024), so a tool may help an author lay out what each claim rests on.

The procedure I propose treats a paper's claims as differing less in kind than in what they rest on. The grounds of a claim may be a cited source, other claims in the paper, a declared starting point such as a definition or a value, or the author's own observation. In the procedure, each kind of grounds gets its own check:

- a **source** is checked in two steps: the work exists, and it says what the claim attributes to it;
- a step from **other claims** is checked by asking whether the warrant holds for the paper's readers;
- a **declared starting point** is checked only for being stated, with a note on whether the field accepts or contests it;
- an **own observation** is checked for a method someone else could repeat.

Combining several sources into a point none of them makes alone adds an inference, which needs the warrant check. A recommendation needs two things more: a declared value it serves, and the conditions under which it applies.

Whetten's criteria work at a different level. He set out what "a complete theory must contain": which factors (What), how they relate (How), why ("the theoretical glue that welds the model together"), and for whom, where and when it holds (Whetten, 1989, pp. 490–492). His seven reviewer questions, from "What's new?" to "Who cares?", answer "What constitutes a publishable theory paper?" (Whetten, 1989, p. 494). They judge a contribution as a whole, so they may serve best as a check of the paper's central contribution, applied once, and not of each statement in it.

These two levels map onto a distinction from systems engineering. Verification shows "proof of compliance with requirements"; validation shows that a product "accomplishes the intended purpose in the intended environment" (NASA, 2016, §2.0). Checking each

claim against its grounds and warrant is verification in this sense; asking Whetten's questions of the contribution is closer to validation. The mapping is mine.

Verification in this sense needs requirements to check against, and the procedure supplies them. A claim meets its requirements when every item it rests on has passed the check for its kind, its warrant holds for the paper's readers, and its qualifier is one that its confidence tier permits. The contribution's requirements are Whetten's elements. The registry then works as a traceability record, of the kind systems engineers keep for requirements: each link, from the contribution to its claims, from a claim to its grounds and from grounds to their sources, is written down, so that someone else can follow and check it. The IPCC asks its authors for a "traceable account" of their evaluation of evidence for the same reason (Mastrandrea et al., 2010).

Qualifiers need a scale. Fixed wording for levels of certainty has precedents. Kent (1964) argued for "a few key odds expressions" in intelligence estimates; the Intergovernmental Panel on Climate Change (IPCC) asks its authors to judge confidence from the "type, amount, quality, and consistency of evidence" and "the degree of agreement" (Mastrandrea et al., 2010); and the GRADE working group (Grading of Recommendations Assessment, Development and Evaluation) fixes wording per level of certainty and size of effect, from "results in" to "may result in" (Santesso et al., 2020). Table 1 adapts this practice with four confidence tiers, each permitting certain qualifiers. It assigns tiers mainly by counting agreeing sources, which is simpler than the IPCC's two axes and has not been tested with readers, as GRADE's wording was. It may still help an author notice when a sentence claims more than its grounds allow.

Table 1: Confidence tiers and the qualifiers they permit.

Tier	**Assigned when...**	**Qualifiers**
Established	Multiple independent sources; no dispute	"shows", "confirms"
Supported	Two or three sources agree; open questions remain	"indicates", "evidence suggests"
Emerging	One or two sources; not replicated	"may", "preliminary evidence"
Speculative	Inference only; no direct evidence	"warrants investigation"

Every claim also carries a *priority*, the author's answer to the question of what fails if the claim is wrong. A P0 claim is one the paper fails without; P1 strengthens the

argument; P2 gives context. The priority sets how complete the checking must be, and every P0 claim is expected to reach the Supported tier.

The central argument of this paper builds on these pieces. If a claim is checked by a procedure that does not fit what it rests on, the check may fail it for the wrong reason: an inference judged by whether it has a source looks unsupported even when its reasoning holds. Recording each claim's grounds item by item, and letting the kind of grounds decide the check, may avoid such false failures. The argument rests on design reasoning and on this paper's own registry; it has not been tested across authors or fields.

4 Limitations

The closest existing practice comes from safety engineering. An assurance case, or safety case, is "a documented body of evidence that provides a convincing and valid argument that a system is adequately safe" for its use; its authors "make an explicit set of claims", link them to the evidence through arguments, and "make clear the assumptions and judgements underlying the arguments" (Bishop and Bloomfield, 1998). The Goal Structuring Notation breaks claims into sub-claims until each can be supported by evidence (Kelly and Weaver, 2004), and the claims–arguments–evidence approach defines its arguments as "Toulmin's warrants" and treats claims asserted without justification as assumptions (Bloomfield and Netkachova, 2014), much as declared starting points work here. The procedure proposed in this paper borrows that discipline for academic writing. It assures the reasoning of a paper where an assurance case assures the safety of a system, and it adds a check of the contribution as a whole.

The vocabulary leaves what is probably the hardest part, judging whether a warrant holds, to the reader. Research on argument quality offers more structure. Wachsmuth et al. (2017) distinguish logical, rhetorical and dialectical quality and define a cogent argument as one with "individually acceptable premises that are relevant to the argument's conclusion and, together, sufficient to draw the conclusion"; Walton et al. (2008) collect argumentation schemes, which their publisher describes as "forms of argument that capture stereotypical patterns of human reasoning". Either could make the warrant check

more specific than Toulmin's layout does.

Whether tools can carry part of the load is also open. Structured prompting was reported to improve LLM argument extraction (Gupta et al., 2024), and representing support and attack formally to improve LLM review aggregation (Sukpanichnant et al., 2024). Both are narrow tasks, so they suggest only that structure combined with LLMs may outperform LLMs alone.

The tier rule is a simplification. Counting sources does not measure their independence or quality, and it gives no guidance for an inference, whose strength lies in its warrant. And the procedure has one user so far: the author of this paper. Verification by someone other than the author would carry more weight. The closest step so far was a review of the first draft by four language models from different vendors, which is independent of the drafting session but not of the method, and no substitute for human reviewers. This commentary is a first step that proposes a way of checking. It does not show that the way works, which needs authors and reviewers in several fields to use it and a comparison with papers checked without it.

5 Recommendations

I propose that the scholarly community develop verification infrastructure for non-empirical papers, as EQUATOR did for empirical health research. The value behind the proposal is that published reasoning should be checkable: readers and reviewers should be able to see what each claim rests on. Its grounds are the gap described in Section 2 and the argument of Section 3. A first version could be modest: authors register the claims a paper depends on, with their grounds, warrants and qualifiers, and publish the register with the paper, as this one does. A first implementation exists: templates for the registry and the glossary, checks for each kind of grounds, and tools that test a registry against its manuscript (see Code and Data Availability).

The proposal applies to academic papers whose contribution is an argument, and to the interpretive parts of empirical papers. The same procedure might serve other writing that asks readers to trust its reasoning, journalism among them, but that is untested.

Toulmin's and Whetten's frameworks come from Western academic traditions; their fit with other scholarly conventions needs to be assessed.

Peer review already judges arguments, and a register would not replace it. It could help authors find weak steps before submission and help reviewers see where a paper's weight rests.

Several directions warrant investigation: testing whether registers change what reviewers find, adapting the checks to the conventions of other fields, building them into writing tools so that the checks run while an author drafts, and recognising the recomputation of derived values as a check of its own.

Acknowledgements and use of AI

The author conceived the idea and developed the framework, starting in March 2026, and made the decisions on scope, structure and the framework's rules. Generative AI was used throughout, under the author's direction: Claude (Anthropic), through Claude Code, drafted and revised the prose, searched for and checked sources, drafted the claim registry and ran the framework's checking tools; the author directed the revisions and reviewed each draft. Four models from different vendors (DeepSeek-V4-Pro, Gemini 3.1 Pro, ChatGPT and Claude) reviewed the first draft and the framework's glossary; their reviews informed this version. The author takes full responsibility for the content. [**Verification record, to be filled from the tool runs of the submitted version, with their date.**]

Code and data availability

The framework described here (templates, glossary and checking tools) and this paper's claim registry are public under a CC BY 4.0 licence at `https://github.com/ducroq/agent-ready-papers`. The version this paper describes is release v5.0.0, archived at [**Zenodo DOI, to be added when the release is archived**].

A Self-Application

This paper was written with the procedure it proposes. Its claim registry, published with the paper,[3] records each claim's grounds, warrant, qualifier, confidence tier and priority, plus the paper's declared starting points and its one contribution. Three entries show how it works.

Table 2 shows three entries side by side: a claim resting on sources, a claim resting on an inference, and the contribution.

Table 2: Three registry entries: what each rests on, how it was checked, and how its priority and confidence tier were set.

	S1-1	**S2-4**	**C1 (contribution)**
Statement	AI citation hallucination is a distinct failure mode.	For non-empirical papers, argument quality is the primary verification challenge.	The community should develop verification infrastructure for non-empirical papers.
Says	descriptive	descriptive	normative
Grounds	two sources: Mugaanyi et al. (2024), Walters and Wilder (2023)	two sources (Toulmin, 2003; Whetten, 1989) and an inference from them	three claims (S2-1, S2-2, S3-4), one declared value (D1), three declared definitions
Check	source check, both steps, for each source	source checks; warrant check, because neither source states the claim	warrant check; boundary conditions and value stated; Whetten's elements (Table 3)
Priority	P0: the opening argument fails without it	P1: strengthens the case for a reasoning check; the paper stands without it	P0 by definition: it is what the paper contributes
Tier	Supported	Emerging	Emerging
How it was set	two peer-reviewed sources agree on the same proposition (Table 1)	the step from the sources to the claim is the author's inference, so the tier cannot exceed Emerging	no higher than its weakest supporting claim (S2-2 and S3-4 are Emerging)
Qualifier in the text	reported findings ("found")	"may"	"I propose"

Under the earlier version of the framework, S2-4 was typed as a sourced claim and its inference was invisible; recording the warrant is what lowered its tier to Emerging.

What the registry shows about this paper. Of the six P0 entries, three reach the Supported tier or above and three do not: the gap claim (S2-2), the central argument (S3-4) and the contribution (C1), which rests on both. By the framework's own rule the

[3] `https://github.com/ducroq/agent-ready-papers/blob/master/papers/perspective/vv/claims/claim_registry.md`

Table 3: Whetten's elements applied to the contribution (C1).

Element	**Content**
What	Verification infrastructure; non-empirical papers; AI-assisted writing (each a declared definition).
How	Non-empirical papers need such infrastructure to stay checkable as AI tools make fluent prose cheap.
Why	Guidelines exist for empirical research (S2-1), none found checks reasoning (S2-2), and each claim needs the checks its grounds call for (S3-4).
Who, where, when	Academic papers whose contribution is an argument, and the interpretive parts of empirical papers; other writing, such as journalism, untested; frameworks from Western traditions.
Value	Published reasoning should be checkable (declared starting point D1).
Objections	Peer review already judges arguments: a register would support review, not replace it.

paper does not pass its gate for critical claims. That is reported here because a framework that asks authors to state the strength of their evidence should state its own.

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